



Effect of P₂O₅ addition on structural and luminescence properties of Nd³⁺-doped tellurite glasses



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ARTICLE INFO

Article history:

Received 13 January 2016

Received in revised form

22 April 2016

Accepted 8 May 2016

Available online 10 May 2016

Keywords:

Phospho-tellurite glasses

Nd³⁺ ion

Raman spectra

Judd-Ofelt parameters

Quantum efficiency

ABSTRACT

Effect of P₂O₅ addition on spectroscopic properties of trivalent neodymium (Nd³⁺)-doped tellurite glasses was studied through Raman, absorption, emission and decay measurements. Raman spectra (350–1250 cm⁻¹) of the Nd³⁺-doped tellurite glasses with various P₂O₅ concentrations, were obtained upon 514 nm Ar⁺ laser excitation and found that bandwidth of the intense band increased with the addition of P₂O₅ from 0 to 20 mol%, indicating the transformation of TeO₄ into TeO₃₊₁ and TeO₃ units. Judd-Ofelt intensity parameters and radiative properties of luminescent ⁴F_{3/2} level were evaluated using absorption spectra for all the glasses. Luminescence spectra were measured in the region 840–1500 nm under 806 nm diode laser excitation. The decay curves for the ⁴F_{3/2} level of Nd³⁺ ion were recorded under 806 nm diode laser excitation by registering the emission at 1058 nm and the lifetime has been found to increase from 188 μs to 249 μs when P₂O₅ concentration increased from 0 to 20 mol%. The figure of merit and quantum efficiency were evaluated and compared to the other reported Nd³⁺-doped glasses. The results indicate that the investigated phospho-tellurite glasses could be useful as laser gain media around 1060 nm.

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1. Introduction

Rare-earth (RE)-doped glasses have been found to be potential candidates for the development of optical devices such as solid state lasers, fiber lasers, fiber amplifiers, up- and down-converting glass layers for solar cells, optical fiber switching, etc. [1–6]. For the past three decades, trivalent neodymium (Nd³⁺)-doped glasses have been largely investigated for solid state lasers since Nd³⁺ ion exhibits prominent emission at around 1060 nm corresponding to ⁴F_{3/2} → ⁴I_{11/2} transition [7,8]. The development of Nd³⁺-doped glasses for laser applications requires hosts with a low maximum phonon frequency to minimize the multiphonon relaxation rates which in turn enhance the quantum efficiency of emitting levels.

Among oxide glasses, TeO₂-based glasses have been attracted as gain media because of their promising properties such as high transmission from UV to IR range, low maximum phonon energy, high linear and nonlinear refractive indices, and good thermal stability and chemical durability [9–11]. Despite a tremendous progress has been made, unfortunately, there were some difficulties such as relatively low thermal stability (158 °C) [12], low tensile strength (2.5–3.2 GPa) [13] and strong up-conversion luminescence of tellurite glasses that have so far been hindering the tellurite glass devices for practical applications. Some of these drawbacks could be overcome by adding a second glass former to the glass matrix such as the P₂O₅ compound which have shown better thermal stability (171–212 °C) [14,15] and tensile strength (4.2–7.2 GPa) [16] values than tellurite glasses. It is expected that this phospho-tellurite glass exhibits a combination of properties from both glass formers. However, adding P₂O₅ to the tellurite matrix could also lead to a change in some interesting optical

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properties of the tellurite matrix such as refractive index, lifetime and quantum efficiency. In addition, the combination of the two glass formers TeO_2 and P_2O_5 is an intrinsically interesting subject of study and are distinct from the properties of either pure tellurite or phosphate networks [17–19].

Hence, the motivation of the present study tries to elucidate the structural and optical properties of Nd^{3+} -doped tellurite glasses with varying P_2O_5 concentration. Raman analysis was carried out to see the effect of P_2O_5 addition on vibrational modes of the tellurite glass network. Judd-Ofelt parameters were determined from the optical absorption spectra and in turn used to evaluate the radiative properties for the $^4\text{F}_{3/2}$ level of Nd^{3+} ions in the present glasses. The lifetime for the $^4\text{F}_{3/2}$ level was obtained from the measured decay curves of the tellurite glasses with varying P_2O_5 concentration.

2. Experimental procedure

The glass samples studied here with the composition: $(75-x)\text{TeO}_2-x\text{P}_2\text{O}_5-6\text{Al}_2\text{O}_3-18\text{K}_2\text{O}-0.5\text{La}_2\text{O}_3-0.5\text{Nd}_2\text{O}_3$, where $x = 0, 10$, and 20 mol%, referred to as TAKLNP0, TAKLNP10, and TAKLNP20, respectively, were prepared by a conventional melt-quenching technique. The batch materials of 20 g were weighed and ground thoroughly in an agate mortar with a pestle for 1 h to obtain homogeneous mixture. The well-mixed powder of all the samples was taken in a platinum crucible and melted in an electric furnace in the range of 900 – 1000 °C for 3 h. Then, the melt was rapidly quenched by pouring it onto a preheated brass plate maintaining in the range of 270 – 290 °C to avoid excess thermal shocks and cracking. After the quenching, the prepared glass samples were immediately transferred to another furnace and subsequently annealed in the range of 270 – 290 °C for 12 h and then slowly cooled down to ambient temperature. The annealing process was made with an objective of minimizing the internal mechanical stress and strain. For optical measurements, the glass samples were cut and polished carefully to get good transparency. The thickness of the glass samples was around 2.38 mm.

Raman spectra of the samples were recorded by Raman microscope (Renishaw inVia) using Ar^+ laser as an excitation source (514 nm). The density of the glasses was measured according to the Archimedes' principle, where the weight of a volume of distilled water equivalent to that of the glass sample was obtained to a precision of 0.0001 g. Linear refractive index measurements (± 0.001) of the glass samples were carried out using a prism coupler system at $\lambda = 632.8$ nm. The UV-VIS-NIR absorption spectra (350 nm– 950 nm) of the glass samples were measured using spectrophotometer (Agilent Technologies) with wavelength resolution of 1.0 nm. The emission spectra were recorded by exciting the samples at 806 nm from a continuous wave laser diode (CWLD). The signal emitted was focused onto a SP-2357 monochromator (Acton Research) and detected by an InGaAs detector (Thorlabs DET10C). The system was controlled with a personal computer where emission spectra were obtained. Special care was taken to maintain the alignment of the setup in order to compare the intensity of the emission signal between different glass samples. The decay time measurements were conducted using a standard setup and coupler impedance with a rise time of 20 μs and comprising a pulsed laser diode at 250 Hz and an InGaAs detector to register the 1058 nm emission.

3. Results and discussion

3.1. Physical and structural properties

Table 1 reports the physical properties of the Nd^{3+} -doped tellurite glasses with varying P_2O_5 concentration. By the addition of

Table 1

Physical properties of the Nd^{3+} -doped tellurite glasses with varying P_2O_5 concentration.

Property	TAKLNP0	TAKLNP10	TAKLNP20
Refractive index	1.8901	1.7872	1.7144
Thickness (cm)	0.238	0.238	0.243
Density (g/cm^3)	4.5362	4.0231	3.6467
Concentration ($\times 10^{20}$ ions/ cm^3)	1.87	1.68	1.54
Dielectric constant	3.57	3.19	2.94
Reflection loss (%)	9.48	7.98	6.93
Interionic distance (Å)	17.14	18.02	18.65
Polaran radius (Å)	7.05	7.30	7.52
Field strength (cm^{-2} , $\times 10^{14}$)	5.98	5.63	5.31

P_2O_5 with respect to TeO_2 into the glass network, the density decreased with an increase in P_2O_5 content. The density values decreased because molecular weight of P_2O_5 (141.94 g/mol) is 0.8 times lower than the molecular weight of TeO_2 (159.6 g/mol).

Fig. 1 shows the normalized Raman spectra of the tellurite glasses with varying P_2O_5 concentration. As shown in Fig. 1, $75\text{TeO}_2-6\text{Al}_2\text{O}_3-18\text{K}_2\text{O}-0.5\text{La}_2\text{O}_3-0.5\text{Nd}_2\text{O}_3$ (TAKLNP0) glass showed a typical Raman spectrum of tellurite glass with major vibrational bands at 465 , 667 and 749 cm^{-1} . The structure of tellurite glass is similar to that of $\alpha\text{-TeO}_2$ [20]. The $\alpha\text{-TeO}_2$ consists of a three dimensional network of infinite chains of TeO_4 trigonal bipyramid (tbp) linked together by shared vertices. Pure TeO_2 hardly forms the glass under normal quenching conditions. The addition of metal oxides to TeO_2 seems to break the axial Te-O bonds because of the strong polarizability of tellurium lone pair electrons and leads to the formation of glass structure. As a result, more non-bridging Te-O bonds are formed and the TeO_4 units are transformed gradually into TeO_{3+1} and TeO_3 type polyhedra when the metal oxides are added more.

The basic structural unit of P_2O_5 glass is the PO_4 tetrahedron, which contains one double-bonded oxygen and three bridging oxygen single bond atoms [21]. The tetrahedral units of phosphate glasses are expressed with the terminology Q_i^i , where 'i' represents the number of bridging oxygens per PO_4 tetrahedron. In addition to the characteristic peaks at 465 , 667 and 749 cm^{-1} , the additional bands at 967 , 1019 and 1129 cm^{-1} were observed due to the P_2O_5 addition in the tellurite glasses. The assignment of the Raman peaks was performed based on the literature available on the tellurite, phosphate, and phospho-tellurite glasses in combination with the

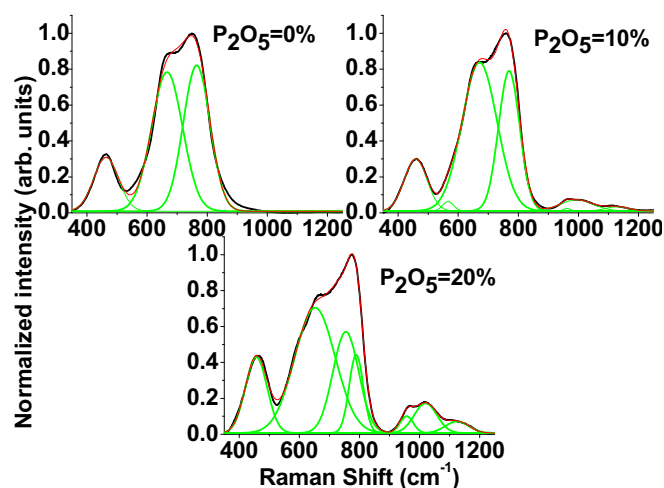


Fig. 1. Raman spectra of the Nd^{3+} -doped tellurite glasses with varying of P_2O_5 concentration.

detailed analysis of various compositions [18–23]. The band at 465 cm^{-1} corresponds to the symmetrical bending and stretching modes of Te-O-P or Te-O-Te linkages. The band at 667 cm^{-1} corresponds to the asymmetric vibrations of the continuous network structure by the connection of TeO_4 trigonal bipyramids and P-O-P bonds. The most intense band at 749 cm^{-1} results from the asymmetric stretching vibrations of P-O-P and stretching vibrations of TeO in TeO_3 trigonal pyramidal units or TeO_{3+1} units.

The presence of Raman bands in the regions of 935–975, 1040–1080 and 1130–1200 cm^{-1} are important in identifying the type Q^i phosphate units available in the glass network. As seen from Fig. 1, the Raman bands at 967, 1019 and 1129 cm^{-1} were observed with the addition of P_2O_5 in tellurite glasses. The Raman bands at 967, 1019 and 1129 cm^{-1} are due to the symmetric stretching mode of non-bridging oxygen atoms (PO_4) in Q^0 phosphate tetrahedra, P-O stretching between Q^1 and Q^0 phosphate tetrahedra, P-O stretching between Q^2 and Q^1 tetrahedra, respectively [21,22]. Intensity of these bands increased with increasing P_2O_5 concentration, which clearly indicating that the addition of P_2O_5 results in the increase of more Q^2 units in the present glass system. The intensity variation of the Raman peaks at 749 cm^{-1} to 667 cm^{-1} was studied in the present glasses with increase of P_2O_5 concentration. It was noticed that with the addition of P_2O_5 from 0 to 20 mol %, the value of intensity ratio increased from 0.12 to 1.28, indicating the transformation of TeO_4 into TeO_{3+1} and TeO_3 . The change can be attributed to two reasons: on the one hand, with substituting TeO_2 by P_2O_5 , will make more Te-O-Te linkages cleaved; on the other hand, P_2O_5 breaks the Te-O-Te linkages, decreases the number of bridging oxygen contents and transforms TeO_4 into TeO_{3+1} and TeO_3 [24,25]. The bandwidth of the major peak at 749 cm^{-1} increased from 186 to 218 cm^{-1} with the addition of P_2O_5 . This bandwidth tuning indicates that the tellurite glass with P_2O_5 content would be promising candidate as a new gain media for broadband Raman fiber amplifier.

3.2. Absorption spectra and Judd-Ofelt analysis

Fig. 2 shows the absorption spectra of the 0.5 mol % Nd_2O_3 -doped tellurite glasses with varying P_2O_5 concentration. As shown in Fig. 2, the absorption spectra exhibited 10 inhomogeneous absorption bands at 431, 470, 515, 527, 586, 629, 685, 749, 805 and 878 nm attributed to $^4I_{9/2} \rightarrow ({}^2P_{1/2}, {}^2D_{5/2}), ({}^2G_{9/2}, {}^2D_{3/2},$

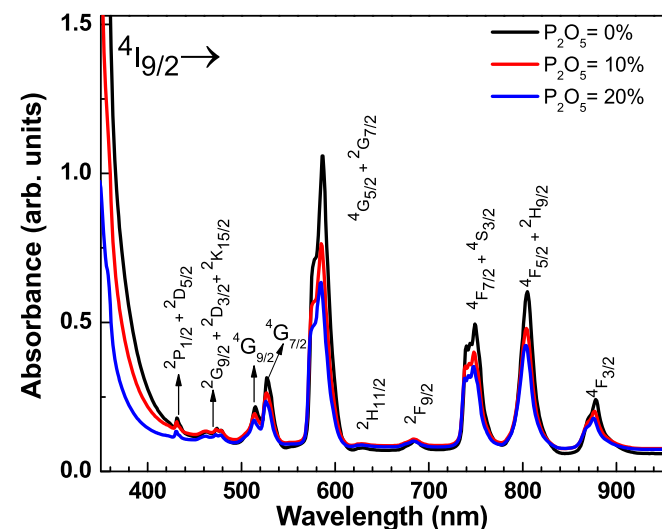


Fig. 2. Optical absorption spectra of Nd^{3+} -doped tellurite glasses with varying P_2O_5 concentration.

${}^2K_{15/2}), {}^4G_{9/2}, {}^4G_{7/2}, ({}^4G_{5/2}, {}^2G_{7/2}), {}^2H_{11/2}, {}^2F_{9/2}, ({}^4F_{7/2}, {}^4S_{3/2}), ({}^4F_{5/2}, {}^2H_{9/2})$ and ${}^4F_{3/2}$ transitions, respectively, obtained due to the 4f-4f transitions of Nd^{3+} ion. The prominent broadband absorption peak locating at 805 nm corresponding to ${}^2H_{9/2} + {}^4F_{5/2}$ transition was observed and the absorption cross-section (σ_{abs}) for this transition was calculated to be 3.13, 3.48, and 3.72 ($\times 10^{-20} \text{ cm}^2$) for 0%, 10% and 20% P_2O_5 concentrations, respectively, which could lead to an efficient energy capture. Therefore, the commercial ~800 nm diode lasers can be adopted as suitable pumping sources for the intense ${}^4F_{3/2} \rightarrow {}^4I_{9/2}, {}^4F_{3/2} \rightarrow {}^4I_{11/2}$ and ${}^4F_{3/2} \rightarrow {}^4I_{13/2}$ emissions, as observed and depicted in Fig. 3.

Judd-Ofelt (JO) theory [26,27] is used to calculate the oscillator strength for the transitions from the ground state to the excited states for the RE ions implanted in different hosts. The experimental oscillator strength (f_{exp}) of each absorption band can be determined by the following expression:

$$f_{\text{exp}} = \frac{mc^2}{\pi e^2 N} \int \alpha(\lambda) d\lambda \quad (1)$$

where 'm' and 'e' are the mass and charge of an electron, 'c' is the velocity of the light in vacuum and 'N' is the Nd^{3+} ion concentration per unit volume, $\alpha(\lambda) = 2.303 D_0(\lambda)/d$ is the absorption coefficient at a given wavelength ' λ ' and $D_0(\lambda)$ is the optical density [$\log I/I_0$].

According to JO theory [26,27], the calculated oscillator strengths (f_{cal}) of an electric-dipole allowed transition $\psi J \rightarrow \psi' J'$ is given as

$$f_{\text{cal}} = \frac{8\pi^2 mc v}{3he^2 (2J+1)} \frac{(n^2+2)^2}{9n} \left(e^2 \sum_{\lambda=2,4,6} \Omega_{\lambda} (\psi J \| U^{\lambda} \| \psi' J')^2 \right) \quad (2)$$

where $e^2 \sum_{\lambda=2,4,6} \Omega_{\lambda} (\psi J \| U^{\lambda} \| \psi' J')^2$ is the line strength of electric-

dipole transitions, 'h' is the Planck's constant, 'v' is the wave-number of the transition in cm^{-1} , 'n' is the refractive index, J and J' are the total angular momenta for the ground and upper levels, respectively, Ω_{λ} ($\lambda = 2, 4$ and 6) are the JO intensity parameters that are characteristic of a given RE^{3+} ion, and $\|U^{\lambda}\|^2$ are the doubly squared reduced matrix elements of the unit tensor operator of the rank $\lambda = 2, 4$ and 6 which are calculated from the intermediate coupling approximation for a transition $\psi J \rightarrow \psi' J'$.

JO intensity parameters, Ω_{λ} and f_{cal} were derived from the f_{exp} by a least-square fitting approach. The matrix elements were taken from the Ref. [28]. The f_{exp} and f_{cal} of TAKLNP glasses were presented in Table 2 along with the Ω_{λ} parameters. As seen from Table 2, the

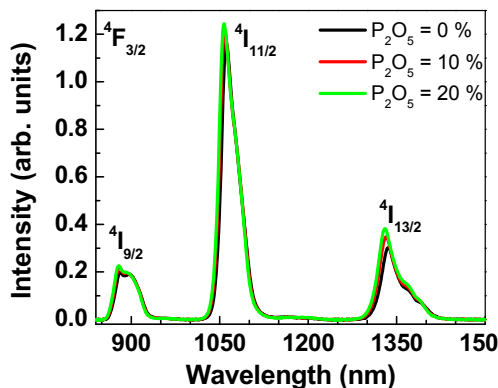


Fig. 3. Emission spectra of the Nd^{3+} -doped tellurite glasses with varying P_2O_5 concentration.

Table 2

Experimental (f_{exp}) and calculated (f_{cal}) oscillator strengths and Judd-Ofelt intensity parameters (Ω_λ , $\times 10^{-20}$ cm²) of absorption bands in the Nd³⁺-doped tellurite glasses with varying P₂O₅ concentration.

Transition $^4I_{9/2} \rightarrow$	TAKLNP0		TAKLNP10		TAKLNP20	
	f_{exp}	f_{cal}	f_{exp}	f_{cal}	f_{exp}	f_{cal}
$^4F_{3/2}$	2.26	2.44	2.03	2.10	2.04	2.25
$^4F_{5/2} + ^2H_{9/2}$	8.07	8.20	7.22	7.32	7.38	7.34
$^4F_{7/2} + ^4S_{3/2}$	8.74	8.70	7.90	7.88	7.63	7.71
$^2F_{9/2}$	0.78	0.66	0.89	0.60	0.58	0.59
$^2H_{11/2}$	0.37	0.18	0.35	0.16	0.19	0.16
$^4G_{5/2} + ^2G_{7/2}$	28.82	28.84	25.03	25.04	23.30	23.31
$^4G_{7/2}$	4.63	4.25	3.71	3.67	3.71	3.69
$^4G_{9/2}$	2.81	1.68	2.49	1.48	2.79	1.52
$^2G_{9/2} + ^2D_{3/2} + ^2K_{15/2}$	1.27	1.16	1.08	1.20	1.18	1.05
$^2P_{1/2} + ^2D_{5/2}$	0.49	0.63	0.30	0.53	0.27	0.59
Ω_2	6.26		5.93		5.47	
Ω_4	3.53		3.23		3.79	
Ω_6	4.74		4.69		4.83	
δ_{rms} (N) ^a	± 0.394 (10)		± 0.350 (10)		± 0.425 (10)	

^a δ_{rms} is the root-mean-square deviation between experimental and calculated oscillator strengths and 'N' refers to the number of levels used in the fitting.

root-mean-square deviation (δ_{rms}) clearly indicates that the fitting process is reliable. As well known, the intensity parameter Ω_2 has been identified to be associated with the asymmetry and the covalency of the RE sites, and Ω_4 and Ω_6 are related to the rigidity of the samples [29]. The decrease in the Ω_2 parameter after adding P₂O₅ is an indication of local structure modification of the glass host, which will be discussed as follows. As TeO₂ and P₂O₅ both are glass formers, there is a distortion of the tellurite network with the addition of the phosphate. The formation of P-O-Te bonds as well as less distorted TeO₆ polyhedra is reported in P₂O₅-TeO₂ glasses with the increase in concentration of P₂O₅ [30]. The obtained Ω_λ parameters were compared to the reported Nd³⁺-doped glasses that include zinc-boro-bismuthate [31] ($\Omega_2 = 2.42$, $\Omega_4 = 3.76$, and $\Omega_6 = 3.45 \times 10^{-20}$ cm²), bismuth borate [32] ($\Omega_2 = 3.28$, $\Omega_4 = 3.88$, and $\Omega_6 = 4.24 \times 10^{-20}$ cm²), and lead fluorosilicate [33] ($\Omega_2 = 3.66$, $\Omega_4 = 1.81$, and $\Omega_6 = 2.10 \times 10^{-20}$ cm²) and found that the present investigated glasses exhibited higher covalency between Nd-O bond and asymmetry in the vicinity of Nd³⁺ ions.

Using Ω_λ parameters and refractive index, several important optical properties, i.e., the radiative transition probability, the branching ratio, and the radiative lifetime were evaluated. The spontaneous emission probability (A) of a transition can be obtained by the following equation:

$$A(\psi'J', \psi J) = \frac{64\pi^4}{3h(2J+1)\lambda^3} \left[\frac{n(n^2+2)^2}{9} S_{\text{ed}} + n^3 S_{\text{md}} \right] \quad (3)$$

where the factor $n(n^2+2)^2/9$ is the local field correction for Nd³⁺ ions in the initial 'J' manifold, ' S_{ed} ' and ' S_{md} ' represent the line strength for the induced electric and magnetic-dipole transitions, respectively, which are expressed by

$$S_{\text{ed}} = e^2 \sum_{\lambda=2,4,6} \Omega_\lambda (\psi J \| U^\lambda \| \psi' J')^2 \quad (4)$$

$$S_{\text{md}} = \left(\frac{e^2 h^2}{16\pi^2 m^2 c^2} \right) (\psi J \| L + 2S \| \psi' J')^2 \quad (5)$$

The total radiative transition probability (A_T) for an excited level is given by the sum of the $A(\psi'J', \psi J)$ terms calculated over all terminal levels,

$$A_T(\psi J) = \sum A(\psi J, \psi' J') \quad (6)$$

The radiative lifetime (τ_{rad}) of an excited level $\psi'J'$ is given by the reciprocal of the $A(\psi J, \psi' J')$,

$$\tau_{\text{rad}}(\psi J) = \frac{1}{A_T(\psi J)} \quad (7)$$

The branching ratio (β_R) corresponding to the emission from an excited level $\psi'J'$ to a lower level ψJ can be calculated from the transition probabilities by using the relation as follows:

$$\beta_R(\psi J, \psi' J') = \frac{A(\psi J, \psi' J')}{A_T(\psi J)} \quad (8)$$

The obtained A , τ_{rad} and β_R for the different transitions of the Nd³⁺ ions were presented in Table 3. It can be seen that the value of the A for all the three emission transitions decreased with increasing P₂O₅ concentration. The β_R was not shown any consistent variation for all the transitions with the increase in P₂O₅ content but the τ_{rad} for the $^4F_{3/2}$ level increased with the increase in P₂O₅ concentration. It was also noticed that the A , τ_{rad} and β_R values for the $^4F_{3/2} \rightarrow ^4I_{11/2}$ transition were higher than those of the other transitions, indicating that this $^4F_{3/2} \rightarrow ^4I_{11/2}$ transition could be useful for laser application around 1.06 μm . The τ_{rad} for the $^4F_{3/2}$ level of present glasses also enlisted in Table 4 along with the reported zinc borate 60Bi₂O₃-39.5(4ZnO-3B₂O₃)-0.5Nd₂O₃ [31], bismuth borate 99.5(40Bi₂O₃-60B₂O₃)-0.5Nd₂O₃ [32], lead fluoro-silicate 67SiO₂-10PbO-2PbF₂-12K₂O-8Na₂O-0.24As₂O₃-0.76Nd₂O₃ [33] for comparison.

3.3. Luminescence spectra

Luminescence spectra of the Nd³⁺-doped tellurite glass samples with varying P₂O₅ concentration are shown in Fig. 3. The emission spectra consisted of three bands centering at 879, 1058, and 1331 nm, which were assigned to the $^4F_{3/2} \rightarrow ^4I_{9/2}$, $^4I_{11/2}$, and $^4I_{13/2}$ transitions of Nd³⁺, respectively. It was noted that the luminescence intensity of the observed transitions slightly increased with increasing P₂O₅ concentration. The effective bandwidth ($\Delta\lambda_{\text{eff}}$) slightly increased for the $^4F_{3/2} \rightarrow ^4I_{11/2}$ transition but no considerable variation was noticed for the rest of the transitions (see Table 3). It was reported that inhomogeneous broadening can result from structural disorder of the glass, which causes differences in ligand electric field at various sites of Nd³⁺ ions depending on the composition of the glass host [34]. The addition of P₂O₅ into TAKLNP glasses changed the local environment of Nd³⁺ ions. The PO₄ units, which participate in the network formation, rupture the Te-O-Te chain in the P₂O₅-TeO₂ glass network [11]. The local ligand fields of the Nd³⁺ ions are influenced by a combination of TeO₃₊₁ units, Te-O-P-groups, PO₄ tetrahedra and the terminal groups of PO₃ and PO₂. The increased bandwidth of $^4F_{3/2} \rightarrow ^4I_{11/2}$ transition in phospho-tellurite glasses is a culmination of two factors, the formation of TeO₃ and TeO₃₊₁ units, the formation of PO₄ tetrahedra, and the PO₃ and PO₂ terminal groups, which also lead to the breaking of tellurite network.

The peak stimulated emission cross-section ($\sigma_e(\lambda_p)$) can be determined using the following equation:

$$\sigma_e(\lambda_p) = \left(\frac{\lambda_p^4}{8\pi c n^2 \Delta\lambda_{\text{eff}}} \right) A(\psi'J', \psi J) \quad (9)$$

where λ_p is the peak wavelength of the emission band and $\Delta\lambda_{\text{eff}}$ is

Table 3Radiative properties and stimulated emission cross-section of the Nd³⁺-doped tellurite glasses with varying P₂O₅ concentration.

Glass label	Transition ⁴ F _{3/2} →	Energy (cm ⁻¹)	A (s ⁻¹)	β _R (%)	Δλ _{eff} (nm)	σ _e (λ _p) (×10 ⁻²⁰ cm ²)	τ _{rad} (μs)
TAKLNP0	⁴ I _{9/2}	11429	1894	38	45	0.94	202
	⁴ I _{11/2}	9570	2511	51	33	3.58	
	⁴ I _{13/2}	7588	517	10	36	1.70	
TAKLNP10	⁴ I _{9/2}	11429	1455	37	45	0.81	256
	⁴ I _{11/2}	9570	2009	51	35	3.00	
	⁴ I _{13/2}	7588	420	11	36	1.52	
TAKLNP20	⁴ I _{9/2}	11429	1432	39	45	0.88	272
	⁴ I _{11/2}	9570	1844	50	35	2.96	
	⁴ I _{13/2}	7588	375	10	37	1.44	

Table 4Stimulated emission cross-section (σ_e(λ_p), ×10⁻²⁰ cm²), experimental (τ_{exp}, μs) and radiative lifetime (τ_{rad}, μs), quantum efficiency (η, %), and figure of merit (σ_e(λ_p) × τ_{exp}, ×10⁻²⁴ cm²s) of ⁴F_{3/2} → ⁴I_{11/2} transition in the Nd³⁺-doped tellurite glasses with varying P₂O₅ concentration along with the reported glasses.

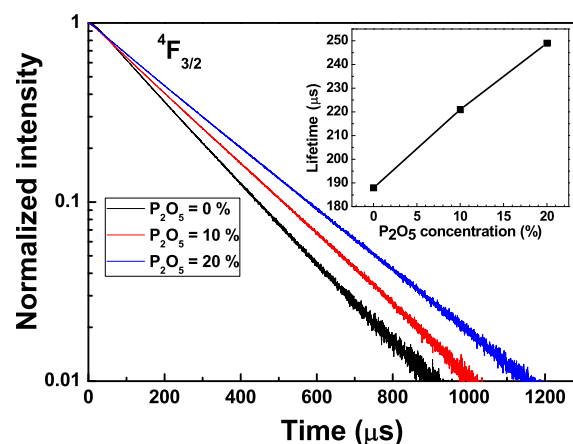
Glass host	σ _e (λ _p) (×10 ⁻²⁰ cm ²)	τ _{exp}	τ _{rad}	η	σ _e (λ _p) × τ _{exp}
TAKLNP0 [This work]	3.58	188	202	93	6.73
TAKLNP10 [This work]	3.00	221	256	86	6.63
TAKLNP20 [This work]	2.96	249	272	91	7.37
Zinc boro-bismuthate [31]	3.95	96	131	73	3.79
Bismuth borate [32]	3.90	95	176	54	3.70
Lead fluoro-silicate [33]	0.87	586	845	69	5.10

its effective bandwidth $\left(\int \frac{I(\lambda)d\lambda}{I_{\max}}\right)$. The σ_e(λ_p) is a very crucial parameter for a laser gain medium. It affects several laser parameters of active medium such as the maximum achievable gain, saturation power, laser threshold, etc. For solid-state four level laser systems, the threshold pump power is inversely proportional to the effective σ_e(λ_p) of the lasing material [35]. In the present glass samples, the σ_e(λ_p) decreased for all the emission transitions with increasing P₂O₅ concentration (see Table 3). The σ_e(λ_p) values for the ⁴F_{3/2} → ⁴I_{11/2} transition of the investigated glasses are tabulated in Table 4 along with the reported glasses for comparison. The present glasses showed higher σ_e(λ_p) values than the reported glasses [31–33] (see Table 4). The product of the emission cross-section and the lifetime of a laser transition is recognized as a figure of merit (σ_e(λ_p) × τ_{exp}) of the laser transition, because σ_e(λ_p) × τ_{exp} is proportional to the slope efficiency and inversely proportional to the threshold pump power of a laser. The σ_e(λ_p) × τ_{exp} for the ⁴F_{3/2} → ⁴I_{11/2} transition was calculated and presented in Table 4 along with the reported glasses [31–33]. It can be seen that the Nd³⁺-doped phospho-tellurite glasses possessed larger figure of merit for 1058 nm emission than the reported glasses [31–33]. Other important properties for the development of fiber laser gain media are non-linear refractive index and high solubility of rare earth. High non-linear refractive index reduces the energy extraction efficiency of the laser and promotes the risk of laser induced damage [36–38]. Also high solubility allow the introduction of large rare earth concentration in a small volume which open the possibility to design smaller laser devices. Since phosphate glasses show low non-linear refractive index and very high solubility of rare earth, it is expected that phospho-tellurite glasses show better properties than tellurite glasses for fiber laser. Hence, the phospho-tellurite glasses are the preferable as laser gain media at 1058 nm.

3.4. Decay curves

Fig. 4 presents the decay curves for the ⁴F_{3/2} level of Nd³⁺ ion in the phospho-tellurite glasses. Decay curves were found to be quite single exponential nature indicating the negligible influence of P₂O₅ concentration even though the concentration of Nd³⁺ ion is

lower. Experimental lifetime (τ_{exp}) was obtained from the first e-folding times of the decay intensity. The τ_{exp} of the ⁴F_{3/2} level of Nd³⁺ ion was found to increase from 188 μs to 249 μs with the increase of P₂O₅ concentration from 0 to 20 mol%. The fluorescence lifetime of Nd³⁺ ion can be described as the decrease in the number of excited ions with time following optical excitation with an infinitesimally short light pulse. As the number of the excited ions is proportional to the fluorescence intensity, integration between t = 0 and t = t yields a single exponential function. In the case of heterogeneous samples, the fluorescence decay could be better described by applying a multiexponential function due to the different mechanism of energy transfer between the ions. However, the decay curves from the samples were found to be quite single exponential nature due to low power excitation of laser. Such fact indicates the negligible influence of different mechanism of energy transfer between the Nd³⁺ ions or additional routes of depopulation from the ⁴F_{3/2} state by non-radiative process. The addition of P₂O₅ to the tellurite glasses may lead to an increase in phonon energy of the glasses since the maximum phonon energy of P₂O₅ is higher than that of TeO₂, which will result in a decrease in lifetime

**Fig. 4.** Decay curves for the ⁴F_{3/2} level of Nd³⁺ ion in tellurite glasses with varying P₂O₅ concentration.

of the $^4F_{3/2}$ level. However, in the present study, the τ_{exp} increased with the increase of P_2O_5 concentration and this may be due to the decrease in the refractive index “n”. According to JO theory, the τ_{rad} is inversely proportional to “n” and thus a decrease of refractive index brings about an increase of the lifetime and the higher symmetry around the Nd^{3+} ion (see inset of Fig. 4). On the other hand, the addition of P_2O_5 to the present tellurite glasses leads to an increase in the phonon energy of the glasses from 749 cm^{-1} to 1129 cm^{-1} which result from the asymmetric stretching vibrations of P-O-P and stretching vibrations of TeO in TeO_3 trigonal pyramidal units; and due to symmetric stretching vibrations in O-P-O groups, respectively. However, the intensity of the Raman band around 749 cm^{-1} is higher than the intensity of the band at 1129 cm^{-1} . The intensity of the Raman bands is proportional to the number of scattering units and their scattering efficiency (Raman cross-section). It may be concluded that the number of O-P-O groups increase as P_2O_5 concentration increases up to 20 mol% but the number of TeO_3 trigonal pyramidal units which are associated with low phonon energy are greater than the O-P-O groups. Thus, the lower the phonon energy of glass host, the larger the phonons needed to bridge the energy gap between the $^4F_{3/2}$ level and the next-lower $^4I_{11/2}$ level, and consequently the lower the probability for non-radiative decay. The experimental results that the τ_{exp} increased with the increase of the maximum phonon energy of the phospho-tellurite glasses clearly indicates that emission of the $^4F_{3/2} \rightarrow ^4I_{11/2}$ transition is more sensitive to effect the symmetry than to the non-radiative processes due to the increase of the maximum phonon energy.

The τ_{exp} for the $^4F_{3/2}$ level of Nd^{3+} ion in the present glasses are tabulated in Table 4 along with the reported glasses for comparison. As seen from Table 4, the τ_{exp} for the $^4F_{3/2}$ level of the present glasses was higher than that of zinc-boro-bismuthate [31] and bismuth-borate [32] glasses but lower than that of lead fluoro-silicate [33] glass. Based on the experimental lifetimes, the quantum efficiency (η) for $^4F_{3/2}$ level can be obtained from the ratio between τ_{exp} and τ_{rad} ($\eta = \tau_{\text{exp}}/\tau_{\text{rad}}$), where τ_{exp} is the experimentally measured lifetime and τ_{rad} is the calculated radiative lifetime. The quantum efficiencies for the $^4F_{3/2}$ level of Nd^{3+} were obtained to be 93%, 86%, and 91%, for TAKLNPO, TAKLNP10, and TAKLNP20 glasses, respectively, and were compared to the other reported glasses [31–33].

4. Conclusion

Nd^{3+} -doped tellurite glasses with varying P_2O_5 concentration were investigated and characterized for their spectroscopic properties. Raman analysis confirmed the presence of phosphate units in the studied tellurite glasses. Judd-Ofelt analysis reveals that the intensity parameters of the Nd^{3+} -doped tellurite glasses decreased with increasing P_2O_5 concentration, showing decrease in the covalent nature of Nd-O bond and increase of symmetry in the vicinity of Nd^{3+} ion. Decay curves for the $^4F_{3/2}$ level of Nd^{3+} ion in the prepared glasses exhibited single exponential and the lifetime was found to increase with increasing P_2O_5 concentration. Figure of merit for the $^4F_{3/2} \rightarrow ^4I_{11/2}$ transition increased with increasing P_2O_5 concentration and it was found to be higher than the reported glasses. Quantum efficiency of the $^4F_{3/2}$ level was found to be between 93% and 86% favoring lasing action at 1058 nm. Hence, it can be concluded that these studies are useful and may be considered

to enhance the laser properties of Nd^{3+} -doped tellurite glasses.

Acknowledgments

The authors would like to thank the Mexican agency CONACYT (No. 164373) and DST, Govt. of India (No. DST/INT/MEXICO/P-102012), for the sanction of Indo-Mexican Joint Research Project. Dr. Ch. Basavapoornima is also grateful to University Grants Commission, New Delhi, for the award of Post Doctoral Fellowship for Women for the year of 2011–12 (F.15-1/2011-12/PDFWM-2011-12-OB-AND-9964 (SA-II), dt. 1-11-2013).

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